Information Theory Class of October 14th 2026

October 13, 2026

Discrete memoryless channel (DMC)



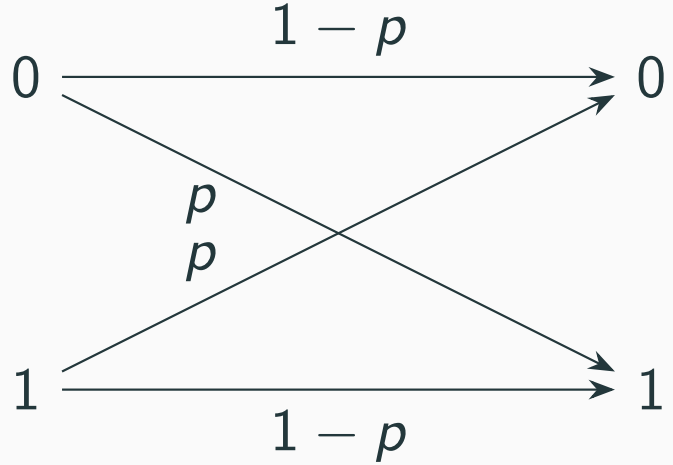
- Finite input alphabet $\mathcal{X}$.
- Finite output alphabet $\mathcal{Y}$.
- Transition law: $Y_t \sim P_{Y|X}(\cdot|x_t)$.

Given the current input, the output does not depend on past inputs or outputs. For a code without feedback,

$$P_{Y^n|X^n}(y^n|x^n) = \prod_{t=1}^{n} P_{Y|X}(y_t|x_t).$$

Binary symmetric channel (BSC)

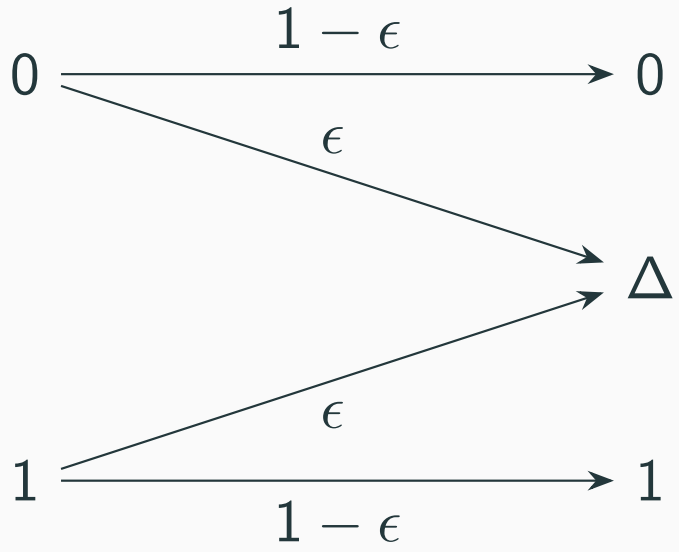
$$\mathcal{X} = \mathcal{Y} = \{0, 1\}.$$



The channel flips each bit with probability p, independently of the other bits.

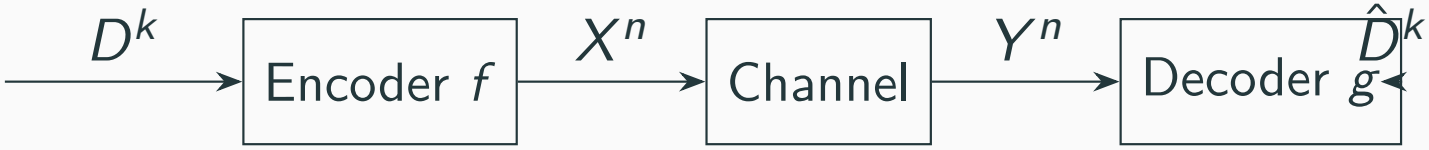
Binary erasure channel (BEC)

$$\mathcal{X} = \{0, 1\}, \qquad \mathcal{Y} = \{0, 1, \Delta\}.$$



The channel erases each transmitted bit with probability ϵ, independently of the other erasures.

Communication over a noisy channel



The rate of a code is

$$R = \frac{k}{n} \qquad \left[\frac{\text{information bits}}{\text{channel use}}\right].$$

The block error probability is

$$P_{\mathrm{e}}^{(n)} = \Pr\{(D_1, \ldots, D_k) \neq (\hat{D}_1, \ldots, \hat{D}_k)\}.$$

We use equiprobable messages and average block error probability.

Achievable rates and capacity

A rate $R > 0$ is achievable if a sequence of codes has rates $R_n \to R$ and

$$P_{\mathrm{e}}^{(n)} \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty.$$

The capacity of a DMC with transition law $P_{Y|X}$ is

$$C = \max_{P_X} I(X;Y).$$

Shannon's channel coding theorem:

1. Every rate $0 < R < C$ is achievable.
2. No rate $R > C$ is achievable.

All logarithms are base 2. Capacity is in bits per channel use.

Achievable rates on a noiseless DMC

Consider the noiseless ternary channel $Y = X$, with $\mathcal{X} = \{0, 1, 2\}$.

- $n = 1$: one information bit can be encoded, so $R = 1$.
- $n = 2$: $k = 3$ bits can be encoded, so $R = 3/2$.
- In general, k bits can be encoded if $2^k \leq 3^n$, or

$$R = \frac{k}{n} \leq \log_2 3.$$

Choose $k = \lfloor n \log_2 3 \rfloor$. The rate tends to $\log_2 3$ with zero error.

$$C = \log_2 3.$$

Computing channel capacity

$$
\begin{aligned}
C &= \max_{P_X} I(X;Y) \\
&= \max_{P_X} \sum_{x,y} P_X(x) P_{Y|X}(y|x) \log_2 \frac{P_{Y|X}(y|x)}{\sum_{x'} P_X(x') P_{Y|X}(y|x')} \\
&= \max_{P_X} [H(Y) - H(Y|X)] \\
&= \max_{P_X} [H(X) - H(X|Y)].
\end{aligned}
$$

Bounds on capacity:

$$C \leq \max_{P_X} \min\{H(X), H(Y)\}$$

$$C \leq \min\{\log_2 |\mathcal{X}|, \log_2 |\mathcal{Y}|\}.$$

For every input distribution P_X, $C \geq I(X;Y)$.

Noiseless channel with m inputs

A noiseless channel maps each of its m inputs to a distinct output.

$$Y = f(X), \quad f \text{ one-to-one}, \qquad H(Y|X) = 0.$$

Therefore,

$$\begin{aligned} C &= \max_{P_X} I(X;Y) \\ &= \max_{P_X} [H(Y) - H(Y|X)] \\ &= \max_{P_X} H(Y) = \log_2 m. \end{aligned}$$

Since $H(Y) \leq \log_2 m$, uniform inputs achieve the maximum by making the m outputs uniform.

BSC capacity: conditional entropy

Define the binary entropy function

$$H_{\mathrm{b}}(p) = -p \log_2 p - (1-p) \log_2 (1-p),$$

with $0 \log_2 0 = 0$.

$$\begin{aligned} H(Y|X=0) &= -(1-p) \log_2 (1-p) - p \log_2 p \\ &= H_{\mathrm{b}}(p). \end{aligned}$$

Similarly, $H(Y|X=1) = H_{\mathrm{b}}(p)$. Thus, for any P_X,

$$\begin{aligned} H(Y|X) &= \sum_{x \in \{0,1\}} P_X(x) H(Y|X=x) \\ &= H_{\mathrm{b}}(p). \end{aligned}$$

BSC capacity

$$C_{\mathrm{BSC}(p)} = \max_{P_X}[H(Y) - H_{\mathrm{b}}(p)] = 1 - H_{\mathrm{b}}(p).$$

Indeed, $H(Y) \leq 1$, and uniform X makes Y uniform. Special cases:

$$C_{\mathrm{BSC}(0)} = C_{\mathrm{BSC}(1)} = 1, \qquad C_{\mathrm{BSC}(1/2)} = 0.$$

When $p = 1/2$, the output is independent of the input, so the channel conveys no information.

BEC capacity: posterior distribution

An unerased output determines the input:

$$H(X|Y=0) = H(X|Y=1) = 0$$

for outputs with positive probability. For the erasure output,

$$P_Y(\Delta) = \sum_{x \in \{0,1\}} P_X(x)\epsilon = \epsilon.$$

If $\epsilon > 0$,

$$P_{X|Y}(x|\Delta) = \frac{P_X(x)\epsilon}{\epsilon} = P_X(x).$$

Observing an erasure does not change the distribution of X.

$$H(X|Y=\Delta) = H(X).$$

BEC capacity

Averaging over the outputs gives

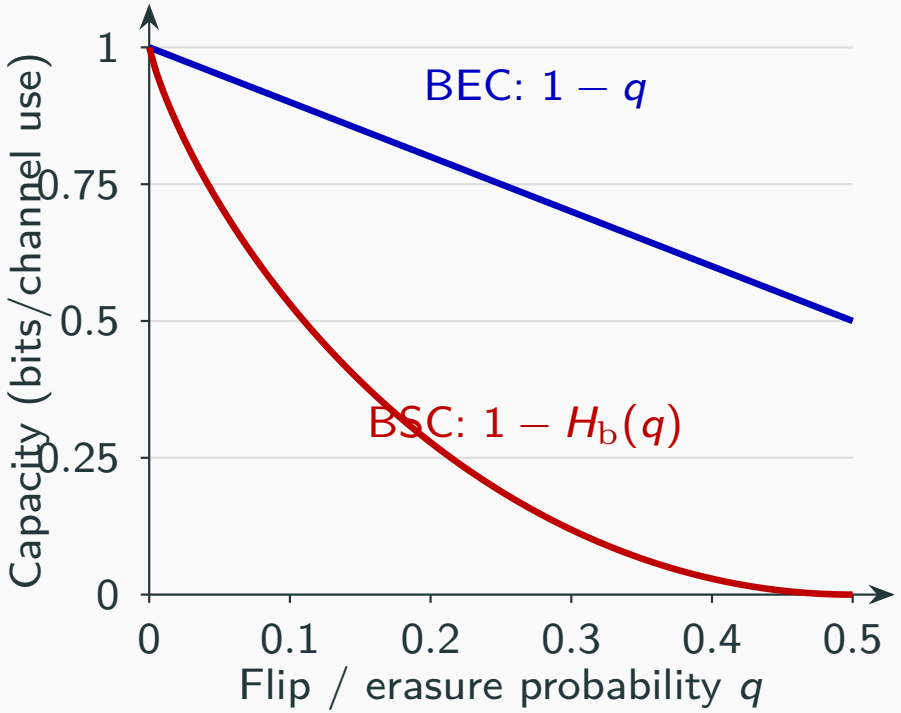
$$\begin{aligned} H(X|Y) &= P_Y(0)H(X|Y=0) + P_Y(1)H(X|Y=1) \\ &\quad + P_Y(\Delta)H(X|Y=\Delta) \\ &= \epsilon H(X). \end{aligned}$$

Consequently,

$$\begin{aligned} C_{\mathrm{BEC}(\epsilon)} &= \max_{P_X}[H(X) - \epsilon H(X)] \\ &= (1-\epsilon)\max_{P_X} H(X) \\ &= 1-\epsilon. \end{aligned}$$

Uniform inputs achieve capacity. The formula also holds at $\epsilon = 0$ and $\epsilon = 1$.

Bit flips and erasures: a capacity comparison



For the same probability $0 < q \leq 1/2$,

$$C_{\mathrm{BEC}(q)} = 1 - q,$$
$$C_{\mathrm{BSC}(q)} = 1 - H_\mathrm{b}(q).$$

Concavity of H_b gives

$$H_\mathrm{b}(q) \geq 2q > q.$$

Therefore,

$$C_{\mathrm{BEC}(q)} > C_{\mathrm{BSC}(q)}.$$

Erasures identify the positions of affected bits. Bit flips do not, so correcting them requires more redundancy.

Gaussian channel capacity

$$Y = X + W, \qquad W \sim \mathcal{N}(0, N_0/2), \qquad W \perp X.$$

For the binary input alphabet $\mathcal{X} = \{-A, A\}$, uniform inputs achieve

$$C_{\text{2-PAM}} = 1 - \frac{1}{\sqrt{\pi N_0}} \int_{-\infty}^{\infty} e^{-u^2/N_0} \cdot \log_2\left(1 + e^{-4A(A+u)/N_0}\right) \, \mathrm{d}u.$$

For $\mathcal{X} = \mathbb{R}$ with the average power constraint $\mathbb{E}[X^2] \le E_s$,

$$C_{\text{Gaussian}} = \frac{1}{2} \log_2\left(1 + \frac{2E_s}{N_0}\right) = \frac{1}{2} \log_2(1 + \text{SNR}).$$

Here $\text{SNR} = 2E_s/N_0$. Capacity is per real channel use.

Converse: code and message notation

Consider a fixed code of block length n for a DMC, without feedback.

- Message M is uniform on $\{1, \ldots, M_n\}$, with $M_n \geq 2$.
- Encoder: $X^n = f_n(M)$.
- Decoder: $\hat{M} = g_n(Y^n)$, taking values in the message set.
- Rate: $R_n = \frac{1}{n} \log_2 M_n$, so $H(M) = nR_n$.
- Average block error: $P_{\mathrm{e}}^{(n)} = \Pr\{\hat{M} \neq M\}$.

For $M_n = 2^k$, this is the earlier notation $M = (D_1, \ldots, D_k)$ and $R_n = k/n$.

$$M \longrightarrow X^n \longrightarrow Y^n \longrightarrow \hat{M}.$$

We will show that $P_{\mathrm{e}}^{(n)} \to 0$ implies $\limsup_n R_n \leq C$.

Fano's inequality

Let M take values in a set of size $M_n \geq 2$, and let $\hat{M} = g_n(Y^n)$. If $P_{\mathrm{e}}^{(n)} = \Pr\{\hat{M} \neq M\}$, then

$$\boxed{H(M|Y^n) \leq H_{\mathrm{b}}(P_{\mathrm{e}}^{(n)}) + P_{\mathrm{e}}^{(n)} \log_2(M_n - 1).}$$

Since $H_{\mathrm{b}}(P_{\mathrm{e}}^{(n)}) \leq 1$ and $\log_2(M_n - 1) \leq nR_n$,

$$\boxed{H(M|Y^n) \leq 1 + P_{\mathrm{e}}^{(n)} nR_n.}$$

Thus, a small decoding error probability limits the uncertainty about the message after observing the output. Fano's inequality itself does not require a uniform message distribution.

Proof of Fano's inequality

Define the error indicator $E = \mathbf{1}\{\hat{M} \neq M\}$. Because E is a function of (M, Y^n),

$$\begin{aligned} H(M|Y^n) &= H(M, E|Y^n) \\ &= H(E|Y^n) + H(M|E, Y^n). \end{aligned}$$

- $H(E|Y^n) \leq H(E) = H_{\mathrm{b}}(P_{\mathrm{e}}^{(n)})$.
- If $E = 0$, $M = g_n(Y^n)$, so the remaining entropy is zero.
- If $E = 1$, at most $M_n - 1$ messages remain possible for each output. The remaining entropy is at most $\log_2(M_n - 1)$.

Hence

$$H(M|E, Y^n) \leq P_{\mathrm{e}}^{(n)} \log_2(M_n - 1),$$

which proves the inequality. Terms corresponding to events of probability zero contribute zero.

Converse, step 1: information and uncertainty

Uniform messages give $H(M) = \log_2 M_n = nR_n$. Split this entropy into

$$nR_n = I(M; Y^n) + H(M|Y^n).$$

The data processing inequality for $M \to X^n \to Y^n$ gives

$$I(M; Y^n) \leq I(X^n; Y^n).$$

Apply Fano's inequality to the second term:

$$\boxed{nR_n \leq I(X^n; Y^n) + 1 + P_{\mathrm{e}}^{(n)} nR_n.}$$

It remains to bound the information that n channel uses can convey.

Converse, step 2: the memoryless channel bound

The entropy chain rule and conditioning reducing entropy imply

$$H(Y^n) = \sum_{t=1}^{n} H(Y_t|Y^{t-1}) \leq \sum_{t=1}^{n} H(Y_t).$$

The channel product law, without feedback, implies

$$H(Y^n|X^n) = \sum_{t=1}^{n} H(Y_t|X_t).$$

Subtracting gives

$$\begin{aligned} I(X^n; Y^n) &= H(Y^n) - H(Y^n|X^n) \\ &\leq \sum_{t=1}^{n} [H(Y_t) - H(Y_t|X_t)] \\ &= \sum_{t=1}^{n} I(X_t; Y_t) \leq nC. \end{aligned}$$

Each $I(X_t; Y_t) \leq C = \max_{P_X} I(X; Y)$. The encoded symbols X_t may be dependent and need not have identical distributions.

Converse, step 3: the rate bound

Combining the previous steps,

$$nR_n \leq nC + 1 + P_{\mathrm{e}}^{(n)} nR_n.$$

Rearrange:

$$\boxed{(1 - P_{\mathrm{e}}^{(n)})R_n \leq C + \frac{1}{n}.}$$

For a reliable sequence of codes, $P_{\mathrm{e}}^{(n)} \to 0$. For sufficiently large n, $P_{\mathrm{e}}^{(n)} < 1$, and

$$R_n \leq \frac{C + 1/n}{1 - P_{\mathrm{e}}^{(n)}}.$$

Taking the limit superior gives

$$\boxed{\limsup_{n \to \infty} R_n \leq C.}$$

Therefore, no rate $R > C$ is achievable.

Converse: a lower bound on the error probability

For $R_n > 0$, the same finite-length inequality yields

$$\boxed{P_{\mathrm{e}}^{(n)} \geq 1 - \frac{C}{R_n} - \frac{1}{nR_n}.}$$

If $R_n \to R > C$, then

$$\liminf_{n\to\infty} P_{\mathrm{e}}^{(n)} \geq 1 - \frac{C}{R} > 0.$$

The error probability stays bounded away from zero above capacity. This proves the *weak converse*, which is sufficient for the stated coding theorem. The *strong converse* asserts $P_{\mathrm{e}}^{(n)} \to 1$ for $R > C$ and requires an additional argument.